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Roll No

EE-302 (GS)
B.E. III Semester Examination, June 2020
Grading System (GS)
Electrical Engineering Materials
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. State the properties and applications of the following high resistivity materials.

- i) Molybdenum
- ii) Platinum
- iii) Tungsten
- iv) Mercury

2. Derive the formula $R_2 = R_1(1 + \alpha_1 t)$, where

R_2 = Resistance at $t_2^{\circ}\text{C}$

R_1 = Resistance at $t_1^{\circ}\text{C}$

t = Temperature rise in $^{\circ}\text{C} = t_2 - t_1$

α_1 = Resistance Coefficient at $t_1^{\circ}\text{C}$

3. Explain about Magnetisation, Magnetic susceptibility coercive force and core temperature.

4. Discuss where the following materials are used in electrical engineering

- i) PVC
- ii) Rubber
- iii) Ceramic
- iv) Bakelite

5. a) Explain in brief fibrous insulating materials and insulating textiles.

b) What are Insulating materials? Explain their electrical mechanical and chemical properties.

OR

What is polarisation? What are types of polarisation? What is polar and non-polar material?

6. a) Write a note on Hard and Soft magnetic materials and draw B-H curve.

b) Write a note on rectangular hysteresis loop.

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7. What is the difference in the band structure of an insulator, a semiconductor, and a conductor? In case of semiconductors indicate the donor and acceptor levels.

OR

- a) Discuss Hall effect.
 - b) Discuss about P-N junction. Discuss the complete characteristic of P-N junction and zener effect.
8. Write short notes on any four of the following
- a) Iron alloys
 - b) Transformer oil properties
 - c) Classification of dielectric
 - d) P and N type impurities
 - e) Properties of mica
